Enhancing Spatial Reasoning of Vision-Language Models on Retail Shelves via Scene Graph

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What?

- **Goal:** answer natural-language questions with accurate spatial reasoning on retail shelf images.
- **Input:** one retail shelf image and one user question.
- **Output:** a natural-language answer supported by visual evidence and spatial relations.
- **Main idea:** combine detection, retrieval support, scene graph, and VLM reasoning, with emphasis on spatial reasoning.

Why?

- Shelf monitoring is still manual, time-consuming, and error-prone.
- Retail shelves are dense and contain many visually similar products, making recognition and reasoning challenging.
- Raw VLMs are weak at counting, localization, and relative-position reasoning (e.g., above, below, left of, near, gap).
- Integrating structured scene graphs with VLMs brings transparency and improves reliability of answers.

Overview

1. Reference Set Preparation

Collect product images and metadata.

2. DINOv2 Embedding

Extract visual features with DINOv2.

3. Vector Database

Store embeddings and metadata.

4. Grounding + Retrieval

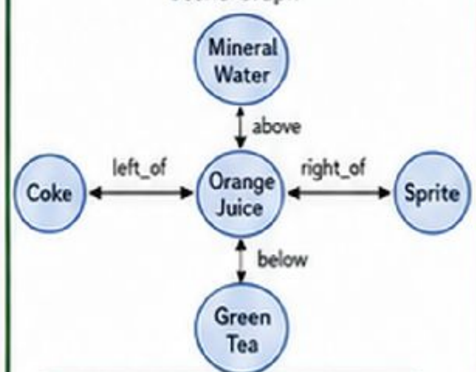
Shelf Image with Detections

Retrieved Candidate Matches

Detect products with Grounding DINO.
Retrieve top-K matches from the vector database.

5. Scene Graph + VLM

Scene Graph



QA Example

Q: Which drink is above the orange juice?
A: The mineral water.

Build a scene graph and use the VLM to answer the question.

Description

1. Research Problem

- Existing retail-shelf datasets rarely support spatial reasoning and natural-language question answering.
- VLMs struggle with counting, localization, and relative-position questions on dense shelves.
- We need a pipeline that grounds products, models spatial relations, and reasons transparently.

Shelf QA Example

Q1: Which drink is above the orange juice?
A1: The mineral water.

Q2: How many Coke bottles are on the middle shelf?
A2: 4.

Q3: Which drink is left of the Sprite?
A3: Coke.

6	128	23	0
Shelves	Products	Product Types	Large Empty Areas

2. Proposed Method

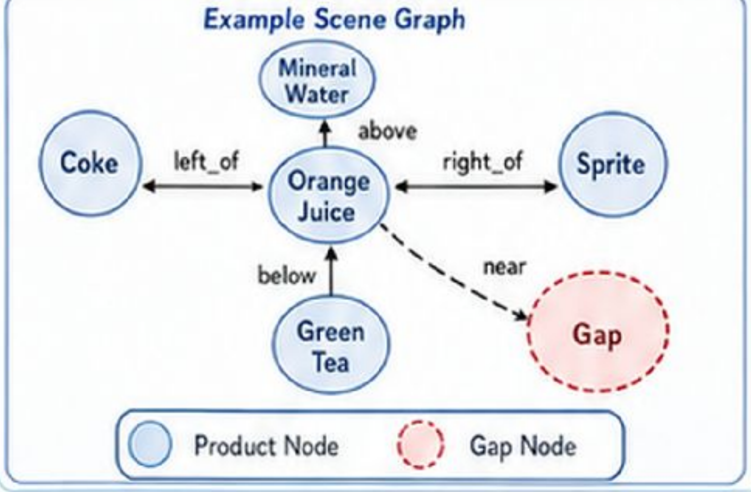
Part 1. Build the Product Vector Database *(Support Preparation)*

- Build a reference image set of products.
- Extract visual features with DINOv2.
- Store embeddings and metadata in a vector database.
- Keep this database to support retrieval in the main pipeline.

Part 2. Shelf Image Understanding and Spatial Reasoning *(Main Focus)*

- Grounding DINO detects product regions in the shelf image.
- Each crop is encoded by DINOv2 and matched to the vector database using cosine similarity.
- Retrieved metadata and bounding-box coordinates are linked and grouped by shelf or product type.
- A scene graph encodes relations such as *left_of*, *right_of*, *above*, *below*, *near*, and *gap*.
- The VLM uses the image, graph, and user prompt to generate the final answer.
- The reasoning process improves interpretability and reduces guesswork.

Example Scene Graph



3. Dataset & Evaluation

- Primary dataset: SKU-110K.
- Original images are the inputs; provided bounding boxes are only ground truth for evaluation.
- Reference set and evaluation set are separated to avoid data leakage.
- Additional subset annotations: retrieved metadata labels, spatial relations, scene-graph labels, and QA pairs.

Evaluation Metrics

Detector	AP50 / AP75, Precision, Recall
DINOv2 + Vector DB	Top-1 Acc., Top-5 Acc., Recall@K, mAP
Scene Graph	Recall@K, Mean Recall@K
VLM Reasoning	Accuracy, F1

Ablation Study (Reasoning Comparison)

1	VLM only
2	VLM + Bounding Boxes
3	VLM + Bounding Boxes + Visual Retrieval
4	VLM + Bounding Boxes + Visual Retrieval + Scene Graph

Expected Outcome

Our pipeline enables stronger spatial reasoning on retail shelf images, improving answer accuracy for relative-position, counting, and gap-related questions with higher interpretability.